ARCHITECTURE SUBMISSION / AUGUST 2026

Skill, Agent, or Neither?

A grounded AI design for five SaaS support capabilities.
Four standalone components. Zero orchestration.

2 SKILLS

2 AGENTS

Prepared as a single-user, provider-agnostic project submission. All customer, policy, pricing, ticket, and changelog data is fictional.

01 / DECISION MAP

Classify the behavior before choosing the artifact

The count is an outcome of one consistent test, not a quota. Deterministic safety controls stay outside the model; stateless procedures become Skills; bounded synthesis or conditional investigation becomes an Agent.

CAPABILITY	DECISION	JUSTIFICATION
New-ticket triage, routing, and immediate escalation	NEITHER	Live-outage, legal, and security escalation must execute as deterministic inbox/on-call rules before any AI queue. Optional classification can follow without owning the safety path.
Tone- and policy-safe outbound replies	SKILL	A repeatable, stateless drafting procedure can load the current policy snapshot, enforce commercial grounding, and return a fixed review block.
At-risk customer one-page briefing	AGENT	The component must reconcile billing, usage, and ticket evidence, label gaps, form hypotheses, and produce a bounded call plan.
Weekly closed-ticket quality grading	SKILL	Rubric application and evidence-backed scoring are stable procedures. Sampling and weekly scheduling are orchestration and are intentionally excluded.
Known-issue check and bug handoff	AGENT	The component performs a conditional investigation: match a changelog entry with evidence, or create a structured new-issue handoff.

Safety principle

An LLM may help categorize a normal ticket, but immediate keyword and outage escalation cannot depend on model latency, availability, or probabilistic recall.

02 / SCOPE MODEL

A narrow component is easier to trust

Each submitted artifact accepts an explicit input and returns a bounded output. None listens to a queue, schedules itself, calls another artifact, mutates customer systems, or sends a message.

1. Deterministic?

Must it execute instantly and identically every time?

Use ordinary rules or software. This is Neither.

2. Stateless procedure?

Does it apply reusable domain guidance to one supplied input?

Package the guidance, references, and validator as a Skill.

3. Bounded investigation?

Must it reconcile sources, branch, or manage uncertainty?

Create a read-only Agent with the minimum input and tool scope.

Explicit boundaries

- No coordinator, end-to-end wiring, event listener, weekly scheduler, or full pipeline demo.
- No backend integrations are required: realistic records are pasted into each standalone workspace.
- No Agent has Memory. No Agent can create a ticket, change an account, send a reply, or approve a commercial term.
- The web app starts in deterministic demo mode and can optionally call a user-selected model.

Provider-agnostic configuration

The single-user Config UI supports OpenAI, Anthropic, Google Gemini, OpenRouter, Ollama, and custom OpenAI-compatible endpoints. Keys remain in browser session storage and requests run only on an explicit click.

03 / STANDALONE SKILL

Policy-safe reply

SKILL

Purpose

Draft a send-ready customer reply in the Acme Cloud voice while grounding every refund, credit, price, discount, eligibility window, and billing commitment in the supplied current policy.

Guardrails

- Treat absent or ambiguous policy as unknown; never estimate or imply a commercial outcome.
- Return two fixed sections: Customer reply and Grounding check.
- Validate monetary tokens in the reply against the supplied policy snapshot.

Realistic test input

```
SUP-1048 / Mia Chen
Growth plan renewed yesterday for USD 149.
Customer asks for a refund; no duplicate charge or billing error.
Policy: renewals are non-refundable except duplicate charge or billing error.
```

Result excerpt

```
Hi Mia,

I checked the renewal details. Subscription renewals are not refundable
unless the charge is duplicated or caused by an Acme Cloud billing error,
and neither applies here.

Grounding check: policy v2026.08.18; monetary claims: none;
human review: not required.
```

PASS

Output shape, tone phrases, and numeric grounding validated by validate_reply.py.

04 / STANDALONE SKILL

Support QA grader

SKILL

Purpose

Apply a weighted quality rubric to closed tickets, cite observable transcript evidence for every deduction, enforce hard caps, and summarize recurring patterns only when the sample supports them.

Rubric

DIMENSION	POINTS	FULL-CREDIT STANDARD
Resolution	30	Resolves the issue or clearly advances it with an owned next step.
Accuracy and policy	25	No unsupported product, refund, pricing, or timeline claim.
Communication	20	Clear, concise, empathetic, and on voice.
Ownership	15	Names the owner and next action.
Process and records	10	Captures material troubleshooting and identifiers.

Realistic test

```
SUP-1082 / failed import E-17
Agent promises an engineering fix by tomorrow and a $200 refund,
then hands off without an owner, diagnostics, or follow-up.
```

Result

```
41 / 100 - Fails standard
Resolution 12/30 | Accuracy 5/25 | Communication 15/20
Ownership 5/15 | Process 4/10
Commercial-claim cap applied: yes
Coaching: collect job ID and logs; avoid unsupported promises; retain ownership.
```

PASS

Score ranges, cap, arithmetic, and evidence presence validated by validate_scorecard.py.

05 / STANDALONE AGENT

Retention brief agent

AGENT

Bounded goal

Turn supplied billing, product usage, and past-ticket records into a one-page pre-call briefing. Reconcile dates, distinguish observations from hypotheses, label missing sources, and never invent a customer motive or commercial offer.

Minimum scope

- Input: one JSON object with account, billing, usage, and ticket sections.
- Tools: none in the mock submission; all records are supplied directly.
- Memory: disabled. Writes, customer contact, and account changes: prohibited.

Realistic test input

```
Northstar Labs / Growth / USD 17,880 ARR / renewal 30 Sep 2026
Billing: latest invoice 12 days late
Usage: WAU 42 -> 17; automation runs 1,280 -> 410
Tickets: SSO issue reopened twice with 2/5 CSAT
```

Result excerpt

```
Risk evidence
- WAU and automation volume fell by more than half.
- INV-883 was paid 12 days late.
- SUP-991 reopened twice and received 2/5 CSAT.

Call plan
1. Confirm new executive and operational owners.
2. Ask which teams stopped using automations and why.
3. Agree on an SSO recovery owner and adoption checkpoint.
```

PASS

All three source sections were traced. Motivations remained labeled hypotheses. No unapproved commercial term was introduced.

06 / STANDALONE AGENT

Bug investigator agent

AGENT

Bounded goal

Compare a customer bug report with the supplied changelog. A known-issue match requires at least two discriminating aligned signals and no material contradiction. Branch to customer guidance for a known issue or an engineering-ready handoff for a new or uncertain issue.

Minimum scope

- Input: one JSON object with report and changelog sections.
- Tools: none in the mock submission; the supplied changelog is the complete search scope.
- Cannot create an engineering ticket or claim a reproduction that was not provided.

Realistic test input

```
SUP-1117: Analytics CSV export stalls at 92% for a 120-day range.
Chrome 127 / EU / web-2026.08.27 / 3 of 3 attempts.
Changelog: Safari PDF stall at 90%; fixed CSV timeout for ranges over one year.
```

Result excerpt

```
Classification: New or unmatched issue.
Title: Analytics CSV export stalls at 92% for ranges over 90 days in EU.
Evidence: KI-224 conflicts on browser and export type; FIX-812 concerns
ranges over one year and is already fixed.
Missing: workspace ID, job ID, timestamps, dataset size, console errors.
```

PASS

The Agent rejected a keyword-only match, preserved uncertainty, and used only supplied steps and evidence.

07 / TEST EVIDENCE

Independent inputs, outputs, and checks

Each artifact was invoked on one realistic mock case without calling any other artifact. The web app exposes the same four independent run surfaces.

CASE	INPUT	EXPECTED BEHAVIOR	RESULT
TC-01 **Policy-safe reply**	Renewal refund request + current policy	No unsupported refund or figure; fixed grounding block	PASS
TC-02 **QA grader**	Closed ticket with invented refund and timeline	Apply score cap; evidence-backed deductions	PASS
TC-03 **Retention brief**	Billing + usage + ticket records	Trace all sources; label hypotheses and confidence	PASS
TC-04 **Bug investigator**	Bug report + two near-match changelog entries	Reject contradictions; produce new-issue handoff	PASS

Validation summary

- Both Skill folders pass the canonical skill structure validator.
- Reply output passes numeric-grounding, required-section, and discouraged-tone checks.
- QA output passes dimension ranges, commercial cap, total arithmetic, and evidence checks.
- The production web build and server-render/API integration tests pass.

4 / 4 standalone cases pass

Mock data is deliberate: backend integrations were not required, and omitting them keeps tool scope honest.

08 / AI USE AND HANDOFF

What AI drafted, what a human decided

AI-assisted work

- Initial classification alternatives and justification drafts.
- First drafts of Skill and Agent instructions, mock fixtures, expected outputs, UI copy, provider adapters, tests, and this report layout.
- Implementation assistance for the single-user web application and PDF generation.

Manual decisions and modifications

- Kept the safety-critical escalation path deterministic and outside the LLM.
- Separated stateless policy/rubric procedure from multi-source and conditional investigation.
- Removed unnecessary tools and Memory from both Agents.
- Defined and reviewed the commercial policy, tone guide, QA weights, hard cap, schemas, evidence threshold, and every expected output.
- Ran structural, arithmetic, build, API, and rendered-PDF verification.

Accessible files

Public source repository

https://github.com/004mayank/saas-support-ai-assignment

Contains Skills, Agents, fixtures, validators, AI-use disclosure, web app source, and submission artifacts.